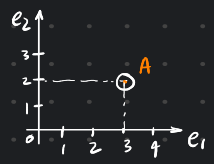
॥ श्रीगणेशाय नमः ॥

Vectors

Points and vectors in 2D

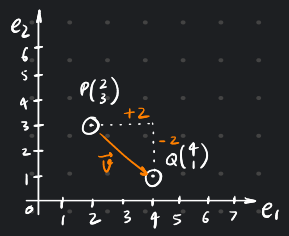
Point: most basic geometric entity that is a particular locⁿ in a coordinate system.

e.g. $A = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$:



$P = \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}$ p_1 units along e_1 axis
p_2 units along e_2 axis.

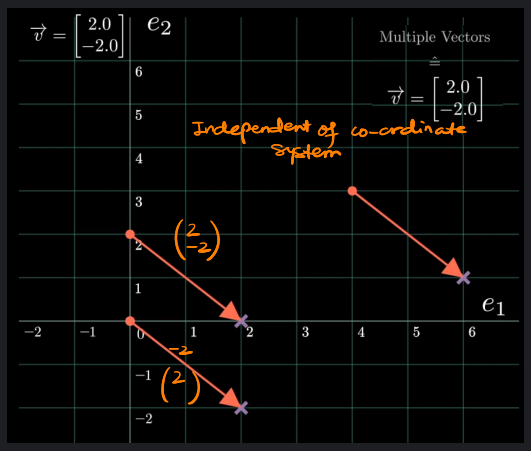
p_1 and p_2 depend on the location of the coordinate origin.



To move from $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ to $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$, we move along the direction of $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$

$P + \vec{v} = Q$

$\vec{v} = Q - P$ $\quad \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} - \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}$



For a point $P\begin{pmatrix} 2 \\ -2 \end{pmatrix}$, the locⁿ in a coordinate system is fixed.
A vector $\vec{v} = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$ can be anywhere in space.

A vector has no fixed position spatially.

Points are co-ordinate system dependent.

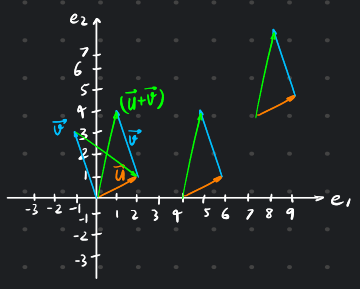
Vectors are co-ordinate independent.

Coordinate independent operations (done on vectors)

- Points $P\begin{pmatrix} p_1 \\ p_2 \end{pmatrix}$, $Q\begin{pmatrix} q_1 \\ q_2 \end{pmatrix}$; $\quad \vec{v} = Q - P = \begin{pmatrix} q_1 - p_1 \\ q_2 - p_2 \end{pmatrix}$
- If we add/subtract 2 vectors, we get another vector.

$\vec{u} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $\quad \vec{v} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

$\vec{u} + \vec{v} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$



We get the same resultant vector regardless of spatial position.

- The resultant of $\vec{u} + \vec{v}$ = diagonal of ||gm formed by $\vec{u}$ and $\vec{v}$.

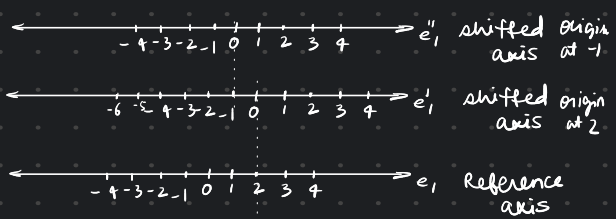
- Adding a vector to a point gives another point.
- Scaling a vector by a constant yields another vector. The scaling adjusts the length of the vector.

$k\vec{v} = k\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} kv_1 \\ kv_2 \end{pmatrix}$

$K > 0$: direction of vector stays the same
$K < 0$: direction is reversed.
$K = 0$: zero vector.

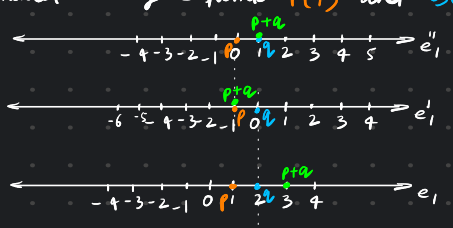
Coordinate dependent operations (on points)

Consider a single-axis system.



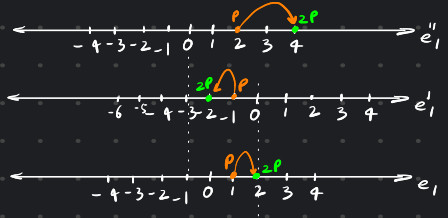
- Addition:

Consider adding 2 points P(1) and Q(2).



NOT well defined.

- Scaling:



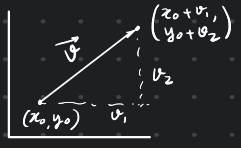
$P = (1)$

$2(P) = ?$

NOT well defined.

Vector length

Say $A = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$ $B = \begin{pmatrix} x_0 + v_1 \\ y_0 + v_2 \end{pmatrix}$. $\vec{v} = B - A = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$



By Pythagoras theorem,

$\|\vec{v}\|^2 = v_1^2 + v_2^2$

Norm of the vector $\vec{v}$ ⇒ length/magnitude of v

$\Rightarrow \|\vec{v}\| = \sqrt{v_1^2 + v_2^2}$ (Euclidean norm)

Suppose we scale $\vec{v}$ by k,

$\|k\vec{v}\| = \sqrt{k^2 v_1^2 + k^2 v_2^2} = |k| \|\vec{v}\|$

$\|k\vec{v}\| = |k| \cdot \|\vec{v}\|$ (length is non-negative)

Unit vector along the direction of $\vec{v}$.

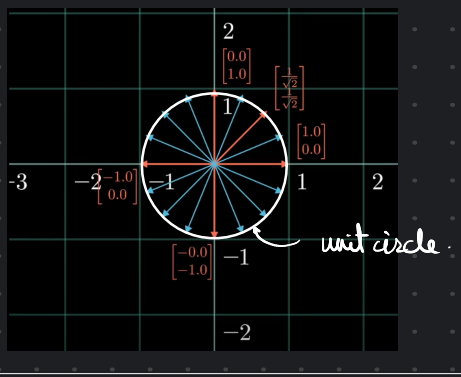
$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$ ← Normalized vector $\|\vec{u}\| = 1$

$\|\vec{v}\| = 0$ iff $\vec{v}$ is the zero vector (all components zero).

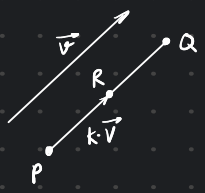
Consider all the unit vectors emenating from the origin in a 2D plane.

$\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix}$, and so on.

We get the unit circle.



Combining vectors



$\vec{v} = Q - P$

$R = (1-k)P + kQ$ is always a point on the line through P and Q.

For $k = 0.5$, R = midpoint of that line.

k and (1-k) are called the coefficients.

Defn: A weighted sum of points, where the coefficients sum up to 1, is called the Barycentric combination.

$R = (1-k)P + kQ$ ← linear interpolation.

Defn: If $0 \le k \le 1$, we call this combination as the convex combination.

E.g. Avg of 2 #s (centre of gravity of points),
expected value of a random variable (probabilities correspond to weights).

To define points outside the line segment PQ, choose $k < 0$ or $k > 1$.



Combining vectors

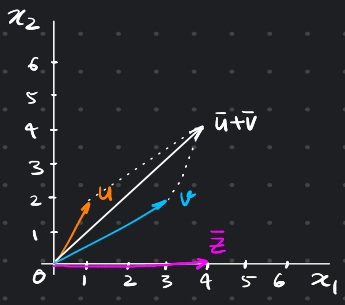
Let u and v be 2 two-component vectors. $\vec{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$, $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$.

$\vec{u} + \vec{v} = \begin{pmatrix} u_1+v_1 \\ u_2+v_2 \end{pmatrix} = \vec{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$ $\quad w_1 = u_1+v_1$, $w_2 = u_2+v_2$ $\quad$ The resultant, w, is another vector.

For some real scalars α_1 and α_2,

$$\alpha_1 u + \alpha_2 v = \begin{pmatrix} \alpha_1 u_1 + \alpha_2 v_1 \\ \alpha_1 u_2 + \alpha_2 v_2 \end{pmatrix} = \vec{z}$$

← linear combination of u and v.

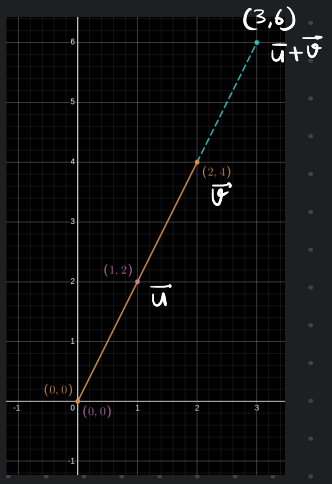


$\vec{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ $\vec{v} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

$\vec{w} = \vec{u}+\vec{v} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$

say $\alpha_1 = 3$, $\alpha_2 = -1$

$\vec{z} = \alpha_1\vec{u} + \alpha_2\vec{v} = \begin{pmatrix} 0 \\ 4 \end{pmatrix}$



Suppose $\vec{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\vec{v} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$. Then $\vec{w} = \vec{u}+\vec{v} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$

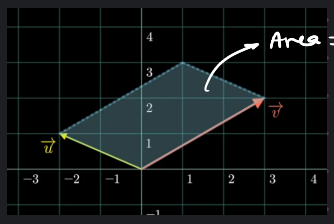
If in 2-D, suppose $\vec{u}$ and $\vec{v}$ are 2 vectors in the same direction (// to each other),

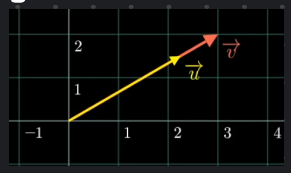
$\vec{u} = \alpha\vec{v}$

we call u and v as linearly dependent vectors.

If $u \neq \alpha v$ for some scalar α, $\vec{u}$ and $\vec{v}$ are linearly independent.

If 2 vectors are LI in 2-D, we say that the llgm formed encloses an area.
If they are LD, then no llgm is formed, area enclosed = 0.

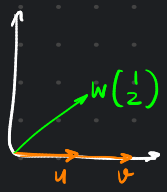




If 2 vectors are LI in 2D, we can reach any destination in 2D, i.e. we can obtain every vector in 2D as a linear combination of these LI vectors.

Any vector $\vec{w} = \alpha_1\vec{u} + \alpha_2\vec{v}$ (α_1, α_2 are scalars, u, v are LI vectors).

Consider an example: $\vec{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\vec{v} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$.



Can we express w as a L.C. of $\vec{u}$ and $\vec{v}$?

$$w \stackrel{?}{=} \alpha_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} \alpha_1 \\ 0 \end{pmatrix} + \begin{pmatrix} 2\alpha_2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} \alpha_1 + 2\alpha_2 \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha_1 + 2\alpha_2 \\ \alpha_1 \cdot 0 + \alpha_2 \cdot 0 \end{pmatrix}$$

we could get α_1 and α_2 to satisfy $\alpha_1 + 2\alpha_2 = 1$.

But we cannot satisfy $\alpha_1 \cdot 0 + \alpha_2 \cdot 0 = 2$ using any α_1, α_2 values.

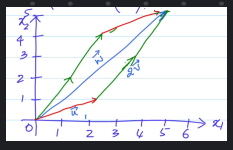
If 2 vectors in 2D are LD, then there's no way to get all the vectors in 2D as a LC of them both. But if they are LI, we can get every vector in 2D as a LC of the 2 vectors.

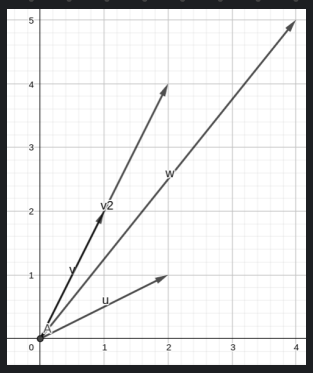
Ex. $\vec{w}=\begin{pmatrix}4\\5\end{pmatrix}$ as a LC of $\vec{u}\begin{pmatrix}2\\1\end{pmatrix}$ and $\vec{v}\begin{pmatrix}1\\2\end{pmatrix}$.

$\rightarrow \begin{pmatrix}4\\5\end{pmatrix} = \alpha_1\begin{pmatrix}2\\1\end{pmatrix} + \alpha_2\begin{pmatrix}1\\2\end{pmatrix} = \begin{pmatrix}2\alpha_1+\alpha_2\\ \alpha_1+2\alpha_2\end{pmatrix}$

$\Rightarrow \begin{pmatrix}4\\5\end{pmatrix} = 1\cdot\begin{pmatrix}2\\1\end{pmatrix} + 2\cdot\begin{pmatrix}1\\2\end{pmatrix}$

$\begin{matrix}2\alpha_1+\alpha_2=4\\ \alpha_1+2\alpha_2=5\end{matrix}\Big\} \alpha_1=1, \alpha_2=2$





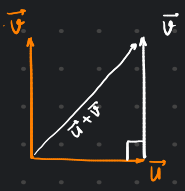
Dot product

Let u and v be 2 vectors in 2D.

① Are u and v orthogonal to each other? ② What is the angle betⁿ them? ③ Are the 2 vectors the same vector?

Case: $\vec{u}$ and $\vec{v}$ are orthogonal.

$\vec{u}=\begin{pmatrix}u_1\\u_2\end{pmatrix}$ $\vec{v}=\begin{pmatrix}v_1\\v_2\end{pmatrix}$



$\|\vec{u}+\vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2$ (Pythagoras theorem)

$(u_1+v_1)^2+(u_2+v_2)^2 = u_1^2+u_2^2+v_1^2+v_2^2$

$2u_1v_1+2u_2v_2=0$

$u_1v_1+u_2v_2=0$

So if $\vec{u}$ and $\vec{v}$ are perpendicular, then the sum of products of components is 0.

$\vec{u}\cdot\vec{v}=0$ where $\vec{u}=\begin{pmatrix}u_1\\u_2\end{pmatrix}$ $\vec{v}=\begin{pmatrix}v_1\\v_2\end{pmatrix}$

↳ $u_1v_1+u_2v_2$ (dot product)

NOTE: If $u=\begin{pmatrix}u_1\\u_2\end{pmatrix}$ is a vector and we want a vector perpendicular to u, then $u_1v_1+u_2v_2=0$

$\Rightarrow \vec{v}=\begin{pmatrix}-u_2\\u_1\end{pmatrix}$

The dot product returns a scalar. (inner product)

$u_{2\times1}$ $v_{2\times1}$ $u^Tv = [\ ]_{1\times2}\begin{bmatrix}\ \\ \ \end{bmatrix}_{2\times1} = [\ ]_{1\times1}$

If $\vec{u}$ and $\vec{v}$ are perpendicular, then $\vec{u}\cdot\vec{v} = u^Tv = v^Tu = \vec{v}\cdot\vec{u} = 0$

How to find the angle betⁿ 2 vectors?

Some properties of dot products

$\vec{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$

- $\vec{u} \cdot \vec{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = u_1^2 + u_2^2 = \|\vec{u}\|^2$. so if $\vec{u} \cdot \vec{u} = 0$, $\vec{u} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

- $\vec{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$, $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$, $\vec{w} = \vec{u} + \vec{v}$

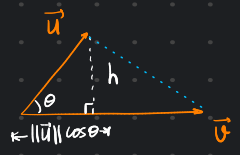
$\vec{w} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$

$$\begin{aligned} \|\vec{w}\|^2 &= w_1^2 + w_2^2 \\ &= (u_1 + v_1)^2 + (u_2 + v_2)^2 \\ &= u_1^2 + 2u_1v_1 + v_1^2 + u_2^2 + 2u_2v_2 + v_2^2 \\ &= u_1^2 + u_2^2 + v_1^2 + v_2^2 + 2(u_1v_1 + u_2v_2) \end{aligned}$$

$$\|\vec{w}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\,\vec{u} \cdot \vec{v}$$

$$\vec{w} \cdot \vec{w} = (\vec{u} + \vec{v})(\vec{u} + \vec{v}) = \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\vec{u} \cdot \vec{v}$$

$$\vec{u} \cdot \vec{u} + \vec{v} \cdot \vec{v} + \vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{u} = \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\vec{u} \cdot \vec{v}$$



$h = \|\vec{u}\| \cos\theta$

$h^2 = \|\vec{u}\|^2 \cos^2\theta$

$h^2 = \|\vec{u}\|^2 (1 - \sin^2\theta)$ —①

By, $h^2 = \|\vec{v} - \vec{u}\|^2 - (\|v\| - \|u\| \cos\theta)$ —②

① & ②, $\|u\|^2 (1 - \sin^2\theta) = \|\vec{v} - \vec{u}\|^2 - (\|\vec{v}\| - \|\vec{u}\| \cos\theta)^2$

$\|\vec{v} - \vec{u}\|^2 = \|\vec{v}\|^2 + \|\vec{u}\|^2 - 2\|\vec{u}\|\,\|\vec{v}\| \cos\theta$ —③ → law of cosines.

$$\begin{aligned} \|\vec{v} - \vec{u}\|^2 &= (\vec{v} - \vec{u})(\vec{v} - \vec{u}) \\ &= \|\vec{v}\|^2 + \|\vec{u}\|^2 - 2\vec{u} \cdot \vec{v} \quad \text{—④} \end{aligned}$$

③ and ④, $\vec{u} \cdot \vec{v} = \|\vec{u}\|\,\|\vec{v}\| \cos\theta$

$$\cos\theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\|\,\|\vec{v}\|}$$

If $\theta = 90°$, $\vec{u} \cdot \vec{v} = 0$.

Comments

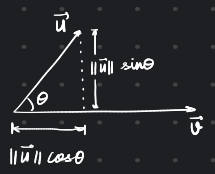
- If $\theta = 90°$, $\cos\theta = 0 \Rightarrow \vec{u} \cdot \vec{v} = 0$
  If θ is acute, $\cos\theta > 0 \Rightarrow \vec{u} \cdot \vec{v} > 0$
  If θ is obtuse, $\cos\theta < 0 \Rightarrow \vec{u} \cdot \vec{v} < 0$.

- If $\theta = 0°$, vectors are in the same direcⁿ (LD).

$\vec{u} \cdot \vec{v} = \|\vec{u}\|\,\|\vec{v}\|$

Orthogonal projections

Projection of a vector $\vec{u}$ in the direction of $\vec{v}$ creates a footprint of length = $||\vec{u}|| \cos\theta$.

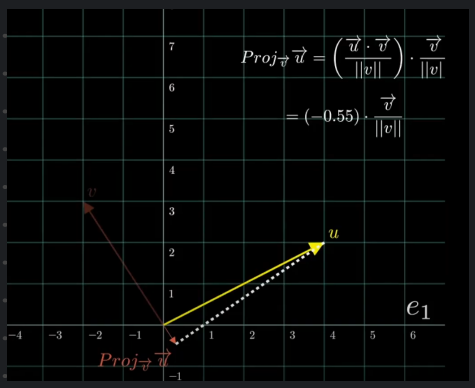


orthogonal projection of $\vec{u}$ onto $\vec{v}$:

$$\text{proj}_{\vec{v}} \vec{u} = \underbrace{\|\vec{u}\| \cos\theta}_{\text{length}} \underbrace{\frac{\vec{v}}{\|\vec{v}\|}}_{\text{unit vector in direction of } \vec{v}}$$

$$= \|\vec{u}\| \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|} \frac{\vec{v}}{\|\vec{v}\|}$$

$$= \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v}$$



The vector $\vec{u}$ can be decomposed into 2 orthogonal components. $\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$

($\cos\theta$ component; $\sin\theta$ component)

$$\vec{u}_{\perp} = \vec{u} - \text{proj}_{\vec{v}} \vec{u} = \vec{u} - \vec{u}_{\parallel}$$

(remove the $\parallel$ comp. to get the $\perp$ comp.)

Lines and parametric equations of lines, linear maps

Lines in 2D: parametric form

Two elements of 2D geometry define a line:

1) 2 points
2) A point and a vector ∥ to the line.
3) A point and a vector ⊥ to the line.

Parametric equation of a line: $l(t) = p + t\vec{v}$

t: parameter (scalar)

$\vec{v} = q - p \Rightarrow l(t) = (1-t)p + tq$

If $0 < t < 1$ → convex combination

$t < 0,\ t > 1$ → extrapolation

E.g. $2x_1 + x_2 = 3$
$4x_1 + 2x_2 = 6$

Homogenous system: $2x_1 + x_2 = 0$, $4x_1 + 2x_2 = 0$ → soln to this lies along the origin and point $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

If HSE has non-trivial soln then it has ∞ly many soln's.

Complete soln to $2x_1 + x_2 = 0$, $4x_1 + 2x_2 = 0$ is given by $x = x_p + k x_H$

x_p: particular soln

x_H: soln to homogeneous system of eq's.

Linear transformations of vectors in 2D

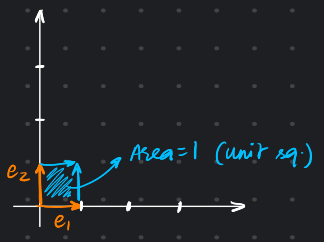
In geometry, we focus on
a) describing objects that can be generated.
b) transforming these objects ←

E.g. consider $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = v_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + v_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = v_1 e_1 + v_2 e_2$$

The vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ get mapped to $\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$.

$\vec{v} = v_1 \vec{e}_1 + v_2 \vec{e}_2$ (linear combination of e_1 and e_2)



We could also express a vector as a lin. comb^n of some other 2 vectors.

Say $\vec{u} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, $\vec{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\vec{w} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$

$\vec{w} = k_1 \vec{u} + k_2 \vec{v}$

We can capture the linear transformation of a vector in matrix form.

$$\begin{bmatrix} 3 \\ 3 \end{bmatrix} = k_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

This transformation matrix maps $k = \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}$ to $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$.

E.g. $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}_{2\times2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2\times1} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{bmatrix}_{2\times1}$ maps $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ to a new vector.

Some linear transformations on vectors in 2D

Scaling a vector

$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \longrightarrow \begin{bmatrix} kv_1 \\ kv_2 \end{bmatrix}$ k: scalar

As a matrix:

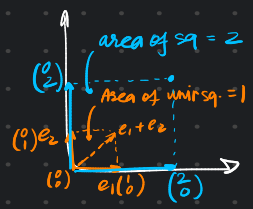
$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} kv_1 \\ kv_2 \end{bmatrix}$$

$a_{11}v_1 + a_{12}v_2 = kv_1 + 0v_2$

$a_{21}v_1 + a_{22}v_2 = kv_2 + 0v_1$

$$\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} kv_1 \\ kv_2 \end{bmatrix}$$

Consider the same vectors $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.



Applying transformation $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ on points of sq.

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 2 & 0 \\ 0 & 0 & 2 & 2 \end{bmatrix}$$

Scaling transformation affects the area of the object that gets scaled.

Reflections

$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \longrightarrow \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

(row swapping)

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_2 \\ v_1 \end{bmatrix}$$

$a_{11}v_1 + a_{12}v_2 = v_2$ (a_{11}: 0, a_{12}: 1)

$a_{21}v_1 + a_{22}v_2 = v_1$ (a_{21}: 1, a_{22}: 0)

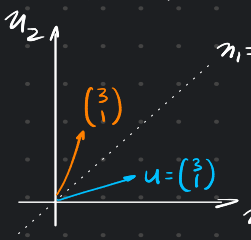
$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_2 \\ v_1 \end{bmatrix}$$

Elementary operation of row swapping is carried out using the matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

To get back to the original vector (undo operation), we transform using the same matrix again.

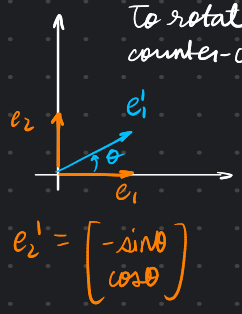
So the inverse of the reflection matrix R is R itself.

The matrix reflects points across the line $x=y$.



$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Rotation



To rotate a vector counter-clockwise by θ.

$e_1' = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$

$e_2' = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$

Rotation matrix:

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1\cos\theta - x_2\sin\theta \\ x_1\sin\theta + x_2\cos\theta \end{bmatrix}$$

$\vec{x}$ → $\vec{x}'$

$$\|\vec{x}\| = \sqrt{x_1^2 + x_2^2}$$

$$\|\vec{x}'\| = \sqrt{(x_1\cos\theta - x_2\sin\theta)^2 + (x_1\sin\theta + x_2\cos\theta)^2}$$

$$= \sqrt{\begin{matrix} x_1^2(\cos^2\theta + \sin^2\theta) + x_2^2(\sin^2\theta + \cos^2\theta) \\ -2x_1x_2\cos\theta\sin\theta + 2x_1x_2\sin\theta\cos\theta \end{matrix}}$$

$$= \sqrt{x_1^2 + x_2^2}$$

$$= \|\vec{x}\|$$

so rotating a vector by an angle θ in 2D preserves the length.

To get $\vec{x}$ back from $\vec{x}'$, we'll rotate counter-clockwise by an angle $-\theta$.

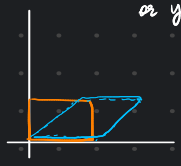
$$R_{-\theta} = \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$R_{-\theta} = (R_\theta)^T$$

Rotation does not change the area.

Shear

$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \longrightarrow \vec{y} = \begin{bmatrix} x_1 + x_2 \\ x_2 \end{bmatrix}$ or $\vec{y} = \begin{bmatrix} x_1 \\ x_1 + x_2 \end{bmatrix}$



$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 \end{bmatrix} \qquad \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} u_1 + ku_2 \\ u_2 \end{bmatrix}$$

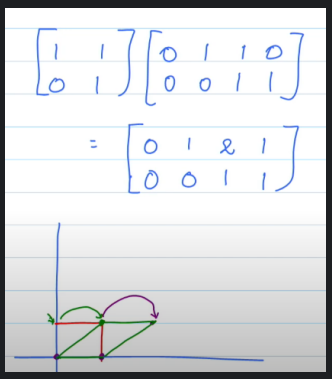
$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_1 + x_2 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 \\ \alpha & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 + \alpha x_1 \end{bmatrix}$$

} Elementary row operation $R_2 \leftarrow R_2 + \alpha R_1$

$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ matrix converts regular font to italic.

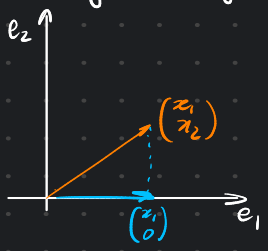
NOTE that the height of the characters stay the same.
Also the feet remain the same.

$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$ are used for elementary row operations.



Projections

Orthogonal projections :

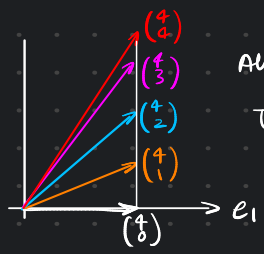


$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \to \begin{pmatrix} x_1 \\ 0 \end{pmatrix}$ projection along the e_1 axis.
(2nd component is 0)

$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \end{bmatrix}$ projection onto e_1 axis

$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \end{bmatrix}$ projection onto the e_2 axis

In all the previous transformations, we could've reversed the effect of the transformations.



All these vectors would have the same projection on the e_1 axis.
There's an information loss of sorts.

So if we say that $\vec{v}$ transformed by $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ is $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$,
we can't judge what $\vec{v}$ actually was.

Projection matrices → NOT invertible
Projection transformations → NOT reversible.

Linear transformations

Suppose A is the matrix representing a linear transformation, then

$$A(\alpha\vec{u} + \beta\vec{v}) = \alpha A\vec{u} + \beta A\vec{v}$$ for some scalars α and β and vectors $\vec{u}$ and $\vec{v}$.

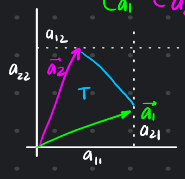
$$A(k\vec{u}) = k(A\vec{u})$$ for some scaling factor k

Checking if a transformation is invertible

Consider a matrix $A^{2\times2}$ as below:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$\vec{a_1}$ $\vec{a_2}$

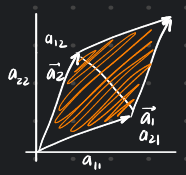


Area of ΔT = Area of complete rectangle − Area of 3 Δs

$$= a_{11}\cdot a_{22} - \left[\frac{1}{2}\times a_{11}a_{21} + \frac{1}{2}a_{12}a_{22} + \frac{1}{2}(a_{11}-a_{12})(a_{22}-a_{21})\right]$$

$$= a_{11}a_{22} - \frac{a_{11}a_{21}}{2} - \frac{a_{12}a_{22}}{2} - \frac{1}{2}(a_{11}a_{22} - a_{12}a_{22} - a_{11}a_{21} + a_{12}a_{21})$$

$$= a_{11}a_{22} - \frac{a_{11}a_{21}}{2} - \frac{a_{12}a_{22}}{2} - \frac{a_{11}a_{22}}{2} + \frac{a_{12}a_{22}}{2} + \frac{a_{11}a_{21}}{2} - \frac{a_{12}a_{21}}{2}$$



Area of Δ $= \frac{a_{11}a_{22}}{2} - \frac{a_{12}a_{21}}{2}$

Area of $\parallel^{gm}$ $= 2\cdot$ Area of Δ

$$= a_{11}a_{22} - a_{12}a_{21}$$

$$= \det(A)$$

For a matrix $A^{2\times2}$, area of $\parallel^{gm}$ made by column vectors of $A = \det(A)$.

① If $\det(A)=1$ → A doesn't change the area.

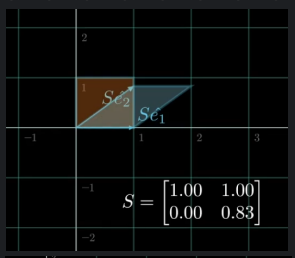
Eg. $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$. $\det(R_\theta)=1$. The rotation matrix doesn't change the area.

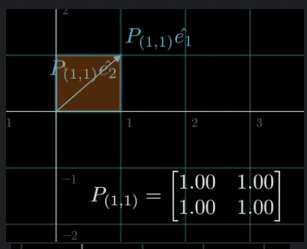
② If $0 \le \det A \le 1$, the linear transformation A shrinks the area.

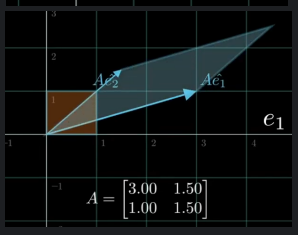
③ If $\det(A)=0$ → rank deficient

④ If $\det(A) > 1$ → Area is expanded.

⑤ If $\det(A) < 0$ → orientation is changed.









Cramer's rule:

$$a_{11}x_1 + a_{12}x_2 = b_1$$
$$a_{21}x_1 + a_{22}x_2 = b_2$$

$$x_1 = \frac{\det\begin{bmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{bmatrix}}{\det\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}} \qquad x_2 = \frac{\det\begin{bmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{bmatrix}}{\det\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}}$$

If $\det A \neq 0$, A is invertible.

For a linear transformation $A^{2\times2}$ to be invertible, $\det A \neq 0$.

$\det(A^{2\times2}) = a_{11}a_{22} - a_{12}a_{21}$. If $\det A = 0 \Rightarrow a_{11}a_{22} = a_{12}a_{21}$

$$\Rightarrow \frac{a_{11}}{a_{21}} = \frac{a_{12}}{a_{22}}$$

⇒ $\det A = 0$ if the col^m vectors are multiples of each other.

No $\parallel^{gm}$ is even formed. col^m vectors are LD.

System of linear eqⁿs:
(2 eqⁿs, 2 unknowns)

$a_{11}x_1 + a_{12}x_2 = b_1$

$a_{21}x_1 + a_{22}x_2 = b_2$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = x_1\begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} + x_2\begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix}$$

$Ax = b$

b is a lin. combinaⁿ of colᵐ vectors of A

When does $A^{2\times2}\,x^{2\times1} = b^{2\times1}$ have i) a unique solⁿ
ii) ∞ly many solⁿs.
iii) no solⁿ ?

System of linear eqⁿs (column/vector perspective)

$2x_1 + x_2 = 3$

$x_1 + 2x_2 = 3$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$\begin{pmatrix} 2 \\ 1 \end{pmatrix} x_1 + \begin{pmatrix} 1 \\ 2 \end{pmatrix} x_2 = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$ ← a lin combⁿ of $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ & $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ gives $\begin{pmatrix} 3 \\ 3 \end{pmatrix}$.

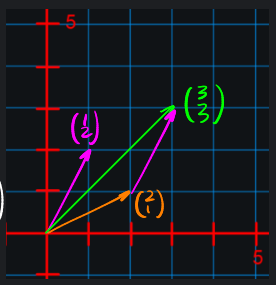
$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} | & | \\ \vec{a}_1 & \vec{a}_2 \\ | & | \end{bmatrix} \qquad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$Ax = b$

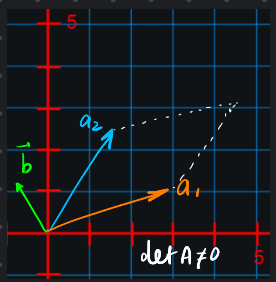
$\vec{b} = x_1\vec{a}_1 + x_2\vec{a}_2$

$1\begin{pmatrix} 2 \\ 1 \end{pmatrix} + 1\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$

soln.



System with unique solⁿ:



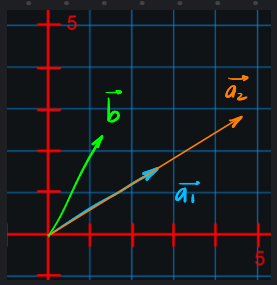
$\vec{b} = x_1\vec{a}_1 + x_2\vec{a}_2$

∞ly many solⁿs:



$\vec{b} = x_1(\vec{a}_1) + x_2(\vec{a}_2)$

no solⁿs:



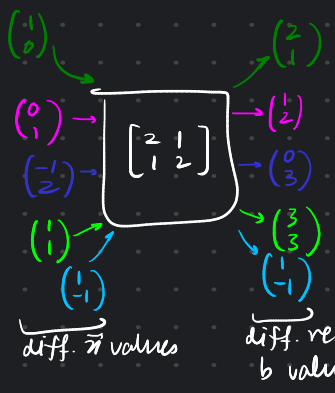
$\vec{b} \neq x_1\vec{a}_1 + x_2\vec{a}_2$

Eigen-pairs

For a transformation matrix A, an eigen-vector of A is a vector $\vec{u}$ s.t. $A\vec{u} = \lambda\vec{u}$,
i.e. A scales the vector $\vec{u}$ without rotating or translating or shearing it.

Say $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

$Ax = b$



} Apart from these 2, the output vectors point in diff. directions than the input vectors.

Scaling factors are 3 and 1.

A vector $\vec{u}$ is called an eigenvector of the matrix A if $A\vec{u} = \lambda\vec{u}$ for some scalar λ.
λ is called the eigenvalue associated with the eigenvector $\vec{u}$.

$$A\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \lambda\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

$A\vec{u} = \lambda\vec{u}$

$\Rightarrow A\vec{u} - \lambda\vec{u} = 0$

$\Rightarrow (A - \lambda I)\vec{u} = 0$

$$\begin{bmatrix} a_{11}-\lambda & a_{12} \\ a_{21} & a_{22}-\lambda \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

← $\vec{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ is the soln to homogeneous system $(A - \lambda I)\vec{u} = 0$

(we want a non-trivial soln)
i.e. $\det(A - \lambda I) = 0$

If $\det A^{2\times 2} = 0$, then the H.S.E. $A\vec{x} = \vec{0}$ has non-trivial soln.

$\det(A - \lambda I) = 0$] → characteristic eqn of A

$$\Rightarrow \det\left(\begin{bmatrix} a_{11}-\lambda & a_{12} \\ a_{21} & a_{22}-\lambda \end{bmatrix}\right) = 0$$

$$\lambda^2 - \underbrace{\lambda(a_{11}+a_{22})}_{\text{sum of diagonal elements called trace}(A)} + \underbrace{(a_{11}a_{22} - a_{12}a_{21})}_{\det A} = 0$$

Product of eigenvalues of A = $\det A$
Sum of eigenvalues of A = Trace (A).

Example:

$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. characteristic eqn $\det\left(\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}\right) = 0$

$\Rightarrow (2-\lambda)^2 - 1 = 0$

$\Rightarrow \lambda^2 - 4\lambda + 3 = 0$

$\Rightarrow \lambda_1 = 3, \lambda_2 = 1$

$\lambda_1 \cdot \lambda_2 = 3 = \det A = 4 - 1$

$\lambda_1 + \lambda_2 = \text{Sum}_{\text{diag.}} A = 2 + 2 = 4$

For $\lambda_1 = 3$: $\begin{pmatrix} 2-3 & 1 \\ 1 & 2-3 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 0$

$\Rightarrow \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 0$

$\left(\begin{array}{cc|c} -1 & 1 & 0 \\ 1 & -1 & 0 \end{array}\right)$

$\left(\begin{array}{cc|c} -1 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right)$

$-1u_1 + 1u_2 = 0$

$u_1 = u_2$

Eigen-vector $= \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

For $\lambda_2 = 1$: $\begin{pmatrix} 2-1 & 1 \\ 1 & 2-1 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\left(\begin{array}{cc|c} 1 & 1 & 0 \\ 1 & 1 & 0 \end{array}\right)$

$u_1 = -u_2$

Eigen vector $= \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Is it always that for a 2x2 matrix, we get real eigenvalues? No, the characteristic equation may not have real solutions.
In that case, the transformation matrix always throws the input vector in some other direction than the original.
Such a matrix is the rotation matrix R^{θ}.

R_θ: rotates every vector by θ.

NOTE: If A is a real symmetric matrix with distinct eigenvalues, the corresponding eigenvectors are orthogonal to each other.